

Coding for Artificial Intelligence

Problem Set 8

Packaging a Module and Automated Style Linting

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Objective

The objective of this practical was to convert the `search4letters` function into a locally installable Python package and use `pycodestyle` to identify and fix PEP 8 style violations.

1 Task 1: Creating the Distribution Package

The package folder was named `text_utility_pkg`. It contained the following files:

```
text_utility_pkg/  
|  
|-- vsearch.py  
|-- setup.py  
|-- README.txt  
|-- dist/  
    |-- vsearch-1.0.tar.gz
```

vsearch.py

```
def search4letters(phrase: str, letters: str) -> set:  
    """Return letters found in phrase."""  
    return set(letters).intersection(set(phrase))  
  
def test_function():  
    """Test function."""  
    pass
```

setup.py

```
from setuptools import setup

setup(
    name='vsearch',
    version='1.0',
    description='Text search tools for AI pre-processing',
    author='Mtech AI Student',
    author_email='student@school.edu',
    url='localhost',
    py_modules=['vsearch'],
)
```

The package was built using:

```
python -m pip install --upgrade build
python -m build --sdist
```

The generated distribution file was:

```
dist/vsearch-1.0.tar.gz
```

It was installed using:

```
python -m pip install .\dist\vsearch-1.0.tar.gz
```

The package import was verified from a directory other than the package directory:

```
import vsearch
vsearch.search4letters('TGGACC', 'GC')
```

Expected result:

```
{'G', 'C'}
```

2 Task 2: Diagnostic Linting and PEP 8 Refactoring

First, pycodestyle was installed using:

```
python -m pip install --upgrade pycodestyle
```

Three intentional PEP 8 errors were introduced into `vsearch.py`.

Before Fixing

The linter was executed using:

```
python -m pycodestyle vsearch.py
```

The detected errors were:

```
E231 missing whitespace after ':'  
W291 trailing whitespace  
E302 expected 2 blank lines, found 1  
W292 no newline at end of file
```

Meaning of Errors

- **E231:** Missing whitespace after a colon in a type hint. A space was added after the colon.
- **W291:** Trailing whitespace at the end of a line. The unnecessary spaces were removed.
- **E302:** Two blank lines are required between top-level functions. The required blank line was added.
- **W292:** The file did not end with a newline. A newline was added at the end of the file.

After Fixing

After correcting all the issues, the command was run again:

```
python -m pycodestyle vsearch.py
```

No output was produced. This indicates that no PEP 8 violations were found and the file passed the style check.

3 Submission Note

The `vsearch` module was successfully converted into a locally installable Python package. After creating `setup.py`, the source distribution was built and the generated `vsearch-1.0.tar.gz` file was installed using `pip`. The import was tested from a directory outside `text_utility_pkg`, and the `search4letters` function returned the expected result. Testing from outside the package directory is important because Python could otherwise find the local `vsearch.py` file directly, which would not prove that the package had actually been installed. For Task 2, three PEP 8 errors were intentionally introduced and detected using `pycodestyle`. The errors were then corrected. A final linting run produced no output, confirming that the cleaned file contained no reported style violations.

Conclusion

This practical demonstrated how to package a Python module, build and install a source distribution, verify the installed package, and use automated PEP 8 linting to identify and correct coding-style problems.